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Author: Oluseye Shay Amusa — Junglenomics FORGE Institute
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PART 1: CHEAT SHEET

Platform at a Glance

Attribute	Value
Platform	AUTODRAFT — Automotive Design Research & AI Fabrication Technology Platform
Document ID	PDD-AUTODRAFT-2026-001
Companion SDD	SDD-AUTODRAFT-2026-001
JCSE	50/50 — ULTRA PREMIUM PLATINUM
Total Prompts	240 Execution Contracts + 27 Governance Prompts = 267 Total
Deployment Phases	5 Phases across 28 Weeks
ZPOS Target	47% Token Reduction 98.2% Semantic Preservation
HIVE Target	97.6/100 — PLATINUM (All 14 Dimensions 90–100%)
AI Models	claude-sonnet-4-6 (primary) · claude-haiku-4-5 (fast ops)
NFT Certification	Polygon L2 · ERC-721 per certified design · IPFS metadata

The Three Engines

Engine	Role	SPC Authority	Core Capability
DYNAMO	Continuous Research Loop	DEXTER ULTRA SI (Seat #9)	8-source crawl · Neo4j graph · frontier detection · push notifications
BLEND	Component Synergy Engine	BAHN × HOLMES ULTRA SI	200K+ components · BOM generator · gap detection · regulatory checker
PIECE MAKER	Gap Manufacturing Engine	BAHN × FRACTAL ULTRA SI (Seat #12)	Generative design · AM process · material selection · FORGE certification

5-Phase Prompt Architecture

#	Phase	Engine / Scope	Prompts	JCSE Gate	SPC Leads
1	Foundation & BAHN Runtime	Infra · Auth · BAHN Service	**50**	40+	ADA ULTRA SI · BAHN ULTRA SI
2	DYNAMO Research Intelligence	Research Loop & Knowledge Graph	**50**	44+	DEXTER ULTRA SI
3	BLEND Component Assembly	Component Library & BOM Engine	**55**	46+	BAHN · HOLMES ULTRA SI
4	PIECE MAKER Gap Manufacturing	Generative Design & AM Pipeline	**58**	48+	BAHN · FRACTAL ULTRA SI
5	Governance, Deploy & Certify	GRO · Infra · HIVE · FORGE	**27**	50/50	ANT KING · ANT QUEEN v2
TOTAL			**240**	**50/50**	**28 Weeks**

Quick Reference

Layer	Value
Frontend	Next.js 14.2 + React 18.3 + TypeScript 5.4 + Tailwind CSS + Three.js r162 (3D)
Backend	NestJS 10.3 + Node.js 20 LTS + BullMQ + Prisma 5.11 + GraphQL (Yoga)
Databases	PostgreSQL 15 · TimescaleDB 2.14 · Neo4j 5.17 · Redis 7.2 · Pinecone · MongoDB 7.0
Infrastructure	Docker 25 · Kubernetes 1.29 (AWS EKS) · Kong API Gateway · Cloudflare Edge
Observability	Prometheus + Grafana · Sentry · Loki · Jaeger
CI/CD	GitHub Actions + ArgoCD (GitOps) + Helm + Terraform
ZPOS Modules	ZPOS · AEOS · NEXUS · PRISM · QUANTUM · SYNTHESIS

PART 2: EXECUTIVE SUMMARY

2.1 Platform Overview

AUTODRAFT is the world's first AI-native automotive design assistant platform, unifying three sovereign intelligence engines — DYNAMO, BLEND, and PIECE MAKER — under a BAHN ULTRA SI domain intelligence layer. This ATLAS PromptWare Design Document translates SDD-AUTODRAFT-2026-001 into 240 discrete, executable prompt contracts organized across five deployment phases. Each prompt is an atomic, AI-implementable instruction mapping directly to a software component defined in the SDD.

The three-engine intelligence flywheel creates self-reinforcing value: DYNAMO discovers → BLEND assembles → PIECE MAKER manufactures → the resulting artifacts re-enter DYNAMO as new manufactured knowledge. The platform grows more intelligent with every design cycle.

2.2 Business Case

Dimension	Value
Market Opportunity	\$2.9T automotive design & engineering industry. Engineers spend 40% of time on information retrieval. 3 deployment scales: Studio / Programme / Enterprise.
Competitive Advantage	Only platform integrating live research, component assembly, and gap manufacturing. BAHN ULTRA SI: 8 automotive domains simultaneously. FORGE + HIVE + NFT certification ecosystem.
Value Proposition	Research-to-BOM in hours, not weeks. Autonomous gap detection and resolution. Real-time regulatory compliance embedded in every assembly.
Revenue Model	Studio: \$800–\$1,200/month · Programme: \$4,000–\$12,000/month · Enterprise: Custom (500+ seats) · NFT certification fees + FORGE premium

2.3 Design Council

SPC Agent	Seat	JCSE	Authority Domain	Primary PDD Phases
ADA ULTRA SI	Seat #1	50/50	System Architecture, Infrastructure, API, Security	Phase 1 (Infrastructure & BAHN) \ Phase 5 (Deployment)
DEXTER ULTRA SI	Seat #9	50/50	Research Architecture, Knowledge Graph, DYNAMO	Phase 2 (DYNAMO Research Intelligence)
BAHN ULTRA SI	Seat #17	50/50	Automotive Domain Intelligence, 8-Domain Routing	Phases 1, 3, 4 (all engines)
FRACTAL ULTRA SI	Seat #12	50/50	Manufacturing Intelligence, Generative Design, AM	Phase 4 (PIECE MAKER)
HOLMES ULTRA SI	Ad hoc	50/50	Deductive Analysis, Synergy Scoring, FMEA	Phase 3 (BLEND Synergy)
STRATEGOS ULTRA SI	Seat #4	50/50	Strategic Intelligence, IP, Competitive Analysis	Phase 5 (Certification)
ANT KING (GRO)	Seat #0	50/50	Royal DNA Governance, LIFE/KILLZONE Mode	Phase 5 (Governance)
ANT QUEEN v2	Colony	50/50	Agent Evolution, Baby Agent Factory, Colony Spawn	Phase 5 (Governance)

2.4 Phase Investment Summary

Phase	Name	Sprints	Prompts	JCSE Gate	Est. Hours
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1	Foundation & BAHN Runtime	S0–S1 (Weeks 1–5)	50	40+	1,000
2	DYNAMO Research Intelligence	S2 (Weeks 6–9)	50	44+	1,000
3	BLEND Component Assembly	S3 (Weeks 10–14)	55	46+	1,100
4	PIECE MAKER Gap Manufacturing	S4 (Weeks 15–19)	58	48+	1,160
5	Governance, Deploy & Certify	S5–S6 (Weeks 20–28)	27	50/50	540
TOTAL		28 Weeks	**240**	**50/50**	**4,800 hrs**

2.5 ZPOS+5 Token Optimisation Strategy

All 240 prompts are subject to ZPOS+5 optimisation before deployment. Target: 47% token reduction, 98.2% semantic preservation. Six modules active across the platform:

Module	Function
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ZPOS	Core token compression via Zipfian frequency patterns
AEOS	Adaptive Engine Optimisation for streaming SSE prompts
NEXUS	Cross-service context linking for multi-engine prompts
PRISM	Multi-modal prompt decomposition for DYNAMO artifacts
QUANTUM	Parallel prompt execution for BLEND synergy scoring
SYNTHESIS	Final compression pass across full prompt pipeline output

PART 3: WORKSHEET

Phase 1 – Foundation & BAHN Runtime (50 Prompts | Sprints 0–1 | Weeks 1–5 | JCSE Gate: 40+)

> Phase 1 establishes the entire platform foundation: cloud infrastructure, authentication, Kong API Gateway, and the BAHN ULTRA SI runtime service. Sprint 0 (Weeks 1–2) delivers infrastructure. Sprint 1 (Weeks 3–5) delivers the BAHN runtime with all 8 domain modes and 7-phase pipeline.

Phase 1A – Infrastructure & Authentication (22 Prompts)

Code	Prompt Specification	SPC Owner	VIBE Tool	JCSE	Status
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INF-001	Scaffold Terraform IaC: AWS EKS 1.29 cluster, 3-node worker group, VPC, subnets, security groups, IAM roles for autodraft-orch, dynamo-svc, blend-svc, piecemaker-svc	ADA ULTRA SI	Cursor IDE	48	

|  Queue |

| ****INF-002**** | Generate Helm chart templates for all 12 AUTODRAFT microservices with environment variable injection, health check endpoints, and rolling update strategy | ADA ULTRA SI | Cursor IDE | 48 |  Queue |

| ****INF-003**** | Configure Kong API Gateway: rate limiting (1000 req/min per user), JWT plugin, CORS, route definitions for all 19 REST endpoints and GraphQL | ADA ULTRA SI | Cursor IDE | 48 |  Queue |

| ****INF-004**** | Set up PostgreSQL 15 RDS schema: users, design_sessions, blend_assemblies, piecemaker_jobs, dynamo_discoveries tables with indexes and constraints | ADA ULTRA SI | Firebase Studio | 47 |  Queue |

| ****INF-005**** | Deploy Redis 7.2 ElastiCache cluster: session_cache and dynamo_queue streams with TTL policies and dead-letter queue for failed crawl jobs | ADA ULTRA SI | Firebase Studio | 47 |  Queue |

| ****INF-006**** | Install and configure TimescaleDB 2.14 extension on PostgreSQL: design_events and research_pulses hypertables with continuous aggregates for dashboards | ADA ULTRA SI | Cursor IDE | 47 |  Queue |

| ****INF-007**** | Configure Prometheus + Grafana observability stack: scrape targets for all 12 services, pre-built dashboards for DYNAMO discovery rate and BLEND assembly latency | ADA ULTRA SI | Firebase Studio | 46 |  Queue |

| ****INF-008**** | Set up GitHub Actions CI/CD pipeline: lint, test, JCSE regression, Docker build, ArgoCD GitOps deployment to EKS with staged rollout gates | ADA ULTRA SI | Cursor IDE | 46 |  Queue |

| ****INF-009**** | Configure Cloudflare Edge: CDN for 3D component previews, ZPOS+5 payload compression at edge, DDoS protection, WebSocket proxying for DYNAMO push | ADA ULTRA SI | Cursor IDE | 46 |  Queue |

| ****INF-010**** | AWS S3 bucket setup: CAD file storage, LCA data archive, PIECE MAKER generative design outputs with per-session AES-256 encryption and versioning | ADA ULTRA SI | Firebase Studio | 46 |  Queue |

| ****AUTH-001**** | Build auth-svc (port 3001): JWT generation (15min access / 7d refresh), OAuth2 authorization code flow, SAML 2.0 for enterprise SSO (Okta, Azure AD) | ADA ULTRA SI | Cursor IDE | 48 |  Queue |

| ****AUTH-002**** | Implement RBAC engine: 5 roles (Designer, Engineer, Senior Engineer, Programme Manager, Admin). Role-gate PIECE MAKER CRITICAL outputs and FORGE certification | ADA ULTRA SI | Cursor IDE | 48 |  Queue |

| ****AUTH-003**** | MFA middleware: TOTP for standard users, hardware key (FIDO2) for enterprise Senior Engineer and Admin roles. Biometric auth support for mobile clients | ADA ULTRA SI | Cursor IDE | 47 |  Queue |

| ****AUTH-004**** | GRO audit logging per session: every authentication event, role escalation, and GRO state transition logged with timestamp, user_id, and session_id | ANT KING | Cursor IDE | 48 |  Queue |

| ****AUTH-005**** | Session isolation architecture: each design session scoped to isolated context. BAHN Royal DNA Protocol activated at session creation, not login | ADA ULTRA SI | Cursor IDE | 48 |  Queue |

| ****AUTH-006**** | EU GDPR compliance: data residency enforcement (eu-west-1 for EU users), PII minimisation in prompt contexts (IDs only, no email/name in AI calls) | ADA ULTRA SI | Cursor IDE | 47 |  Queue |

| ****AUTH-007**** | EU AI Act High-Risk classification audit log: human oversight record for PIECE MAKER CRITICAL components, GRO HUMAN_IN_LOOP transition records | ADA ULTRA SI | Cursor IDE | 47 |  Queue |

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| **GW-001** | Kong route definition: /api/session/*, /api/bahn/*, /api/dynamo/*, /api/blend/*, /api/piecemaker/*, /api/forge/*, /api/analytics/* with service upstream configs | ADA ULTRA SI | Cursor IDE | 46 |  Queue |

| **GW-002** | ZPOS+5 payload compression plugin: intercept all Claude API responses before delivery, apply SYNTHESIS pass, inject semantic_preservation_score header | ADA ULTRA SI | Cursor IDE | 47 |  Queue |

| **GW-003** | WebSocket manager: persistent connection for DYNAMO push notifications. Route CRITICAL priority alerts immediately, queue LOW priority in 5-minute batches | ADA ULTRA SI | Cursor IDE | 46 |  Queue |

| **GW-004** | Streaming SSE proxy: /api/bahn/fullbuild and /api/piecemaker/design/stream endpoints. 30-minute connection timeout, chunked transfer encoding, reconnection logic | ADA ULTRA SI | Cursor IDE | 46 |  Queue |

| **GW-005** | API Gateway health dashboard: real-time error rates per route, p95 latency per engine, rate limit hit frequency, ZPOS compression ratio metrics | ADA ULTRA SI | Firebase Studio | 45 |  Queue |

Phase 1B – BAHN ULTRA SI Runtime Service (28 Prompts)

| Code | Prompt Specification | SPC Owner | VIBE Tool | JCSE | Status |

|---|---|---|---|---|---|

| **BAHN-001** | Deploy bahn-runtime service (port 3040): NestJS module structure for BAHN ULTRA SI – 8-domain router, 7-phase pipeline orchestrator, mode switch controller | BAHN ULTRA SI | Cursor IDE | 50 |  Queue |

| **BAHN-002** | Implement BAHN:CONCEPT mode: session creation handler, design DNA extraction from vehicle spec, sustainability target parsing, regulatory scope selection | BAHN ULTRA SI | claude-sonnet-4-6 | 50 |  Queue |

| **BAHN-003** | Implement BAHN:ENGINEER mode: chassis/BIW/suspension constraint definition, load path validation schema, crashworthiness threshold enforcement (Euro NCAP) | BAHN ULTRA SI | claude-sonnet-4-6 | 50 |  Queue |

| **BAHN-004** | Implement BAHN:MATERIALS mode: MCS tool integration – yield strength, fatigue, thermal checks; REACH compliance; AM process compatibility filtering | BAHN ULTRA SI | claude-sonnet-4-6 | 50 |  Queue |

| **BAHN-005** | Implement BAHN:POWERTRAIN mode: APDM decision matrix – BEV/FCEV/PHEV/H2-ICE routing, battery architecture templates, thermal management requirements output | BAHN ULTRA SI | claude-sonnet-4-6 | 50 |  Queue |

| **BAHN-006** | Implement BAHN:COMPLY mode: multi-market regulatory matrix (EU/US/CN/IN/ROW), type approval pathway generator, UNECE/NHTSA/EU compliance checker | BAHN ULTRA SI | claude-sonnet-4-6 | 50 |  Queue |

| **BAHN-007** | Implement BAHN:MANUFACTURE mode: production floor economics calculator, FRACTAL cell design integration, assembly process optimisation, LCA closure report | BAHN ULTRA SI | claude-sonnet-4-6 | 50 |  Queue |

| ****BAHN-008**** | Implement BAHN:SOVEREIGN mode (enterprise only): maximum depth activation across all 8 domains simultaneously, FULLBUILD session handler, critical design review format | BAHN ULTRA SI | claude-sonnet-4-6 | 50 |  Queue |

| ****BAHN-009**** | D1 Automotive Conceptual Design domain router: design DNA parser, character line logic, proportional analysis, surfacing/design intent outputs to BLEND | BAHN ULTRA SI | claude-sonnet-4-6 | 49 |  Queue |

| ****BAHN-010**** | D2 Mechanical & Systems Engineering domain router: topology optimisation trigger, suspension geometry, crashworthiness module; structural gap signal to PIECE MAKER | BAHN ULTRA SI | claude-sonnet-4-6 | 49 |  Queue |

| ****BAHN-011**** | D3 Sustainable Manufacturing domain router: AM process selector trigger, FRACTAL factory cell design, PIECE MAKER process guidance output | BAHN ULTRA SI | claude-sonnet-4-6 | 49 |  Queue |

| ****BAHN-012**** | D4 Alternative Powertrain Intelligence domain router: APDM BEV/FCEV/PHEV/H2-ICE classification, battery architecture mapping to BLEND component categories | BAHN ULTRA SI | claude-sonnet-4-6 | 49 |  Queue |

| ****BAHN-013**** | D5 Materials Science domain router: MCS CO2e/kg scoring, bio-composite and recycled polymer candidate lists, fatigue data injection to PIECE MAKER design brief | BAHN ULTRA SI | claude-sonnet-4-6 | 49 |  Queue |

| ****BAHN-014**** | D6 Regulatory & Homologation domain router: Phase 6 pipeline activation, multi-market compliance matrix builder, safety flag taxonomy, REACH substance check trigger | BAHN ULTRA SI | claude-sonnet-4-6 | 49 |  Queue |

| ****BAHN-015**** | D7 Industrial Design & Human Factors domain router: HMI architecture, ergonomics constraints, ADAS interface specs, accessibility requirements to BLEND interior library | BAHN ULTRA SI | claude-sonnet-4-6 | 49 |  Queue |

| ****BAHN-016**** | D8 LCA & Circular Economy domain router: ISO 14040/44 LCA trigger, EPD generation, supply chain carbon allocation, end-of-life recovery target enforcement | BAHN ULTRA SI | claude-sonnet-4-6 | 49 |  Queue |

| ****BAHN-017**** | Phase 1 (Concept Genesis) pipeline: design brief intake, market positioning analysis, sustainability target setting, BAHN:CONCEPT mode output structure | BAHN ULTRA SI | claude-sonnet-4-6 | 50 |  Queue |

| ****BAHN-018**** | Phase 2 (Proportional Engineering) pipeline: proportion targets, package envelope, human factors baseline, character line logic output to BLEND D1 constraints | BAHN ULTRA SI | claude-sonnet-4-6 | 49 |  Queue |

| ****BAHN-019**** | Phase 3 (Systems Architecture) pipeline: BIW load path, suspension hardpoints, powertrain packaging, BAHN:ENGINEER constraint set export to BLEND | BAHN ULTRA SI | claude-sonnet-4-6 | 49 |  Queue |

| ****BAHN-020**** | Phase 4 (Advanced Technology Integration) pipeline: materials shortlist, AM process candidates, sustainable manufacturing assessment, PIECE MAKER material brief | BAHN ULTRA SI | claude-sonnet-4-6 | 49 |  Queue |

| ****BAHN-021**** | Phase 5 (Powertrain Integration) pipeline: APDM matrix execution, battery thermal management, charging architecture, BLEND powertrain component scope | BAHN ULTRA SI | claude-sonnet-4-6 | 49 |  Queue |

| ****BAHN-022**** | Phase 6 (Regulatory Homologation) pipeline: multi-market compliance check, type approval pathway, safety certification map, REACH substance validation | BAHN ULTRA SI | claude-sonnet-4-6 | 49 |  Queue |

| ****BAHN-023**** | Phase 7 (Production Engineering) pipeline: FRACTAL factory integration, LCA closure, production cost analysis, PIECE MAKER AM process output formatting | BAHN ULTRA SI | claude-sonnet-4-6 | 49 |  Queue |

| ****BAHN-024**** | ZPOS+5 integration for BAHN runtime: pre-compression of BAHN domain prompt cores to ULTRA SI specification. Target 47% reduction, 98.2% semantic preservation | ADA ULTRA SI | claude-sonnet-4-6 | 48 |  Queue |

| ****BAHN-025**** | SSE streaming handler for BAHN ULTRA SI responses: 7-phase pipeline streaming with domain-level progress events, error recovery, and reconnection management | ADA ULTRA SI | Cursor IDE | 47 |  Queue |

| ****BAHN-026**** | BAHN health monitoring: response time SLA per mode (<3s CONCEPT, <8s ENGINEER, <15s SOVEREIGN), degradation alerts, auto-fallback to CONCEPT | ADA ULTRA SI | Firebase Studio | 46 |  Queue |

| ****BAHN-027**** | BAHN Royal DNA embed: GRO mode switch logic integrated at bahn-runtime layer — KILLZONE trigger for weapons/combat vehicle requests, SAFE_LIFE for safety recalls | ANT KING | claude-sonnet-4-6 | 50 |  Queue |

| ****BAHN-028**** | BAHN automotive cybersecurity compliance: UN ECE R155 cybersecurity management checks for connected vehicle data flows, R156 OTA compliance validation | BAHN ULTRA SI | Cursor IDE | 47 |  Queue |

Phase 2 — DYNAMO Research Intelligence (50 Prompts | Sprint 2 | Weeks 6–9 | JCSE Gate: 44+)

> Phase 2 builds the DYNAMO Continuous Research Loop Engine — the perpetually active intelligence discovery system. DEXTER ULTRA SI architects the 8-source crawl pipeline, NLP artifact parser, relevance scorer, frontier zone detector, and all Neo4j knowledge graph schemas.

Phase 2A — DYNAMO Core Engine (32 Prompts)

| Code | Prompt Specification | SPC Owner | VIBE Tool | JCSE | Status |

|---|---|---|---|---|---|

| ****DYN-001**** | Build dynamo-svc (port 3010): NestJS module with BullMQ job queues for 8 crawl workers, cron scheduler with configurable frequencies per source domain | DEXTER ULTRA SI | Cursor IDE | 50 |  Queue |

| ****DYN-002**** | USPTO/EPO patent feed crawler: RSS + API polling every 6 hours, deduplication by patent number hash, automotive keyword filter (IPC class B60, B62) | DEXTER ULTRA SI | Cursor IDE | 49 |  Queue |

| ****DYN-003**** | PubMed/IEEE/SAE peer-reviewed paper crawler: DOI-based deduplication, 12-hour cycle, citation count harvesting, automotive/materials/powertrain filter | DEXTER ULTRA SI | Cursor IDE | 49 |  Queue |

| ****DYN-004**** | UNECE/NHTSA/EU regulatory RSS crawler: 24-hour cycle, regulation identifier extraction, effective date parsing, affected vehicle category tagging | DEXTER ULTRA SI | Cursor IDE | 49 |  Queue |

| ****DYN-005**** | Tier 1/2 supplier technical bulletin crawler: 4-hour cycle, supplier-specific API connectors (Bosch, Continental, Magna APIs), spec update detection | DEXTER ULTRA SI | Cursor IDE | 48 |  Queue |

| ****DYN-006**** | Materials database crawler: CES EduPack and Granta Design API integration, 8-hour cycle, CO2e/kg, fatigue, and thermal property updates | DEXTER ULTRA SI | Cursor IDE | 48 |  Queue |

| ****DYN-007**** | Additive manufacturing process database crawler: ExOne, EOS, Stratasys API feeds, 12-hour cycle, new material/parameter/process capability harvesting | DEXTER ULTRA SI | Cursor IDE | 48 |  Queue |

| ****DYN-008**** | Design trend and CAE software release crawler: Pininfarina/Giugiaro trend feeds + ANSYS/Abaqus/Altair release notes, 24–48 hour cycle, automotive relevance filter | DEXTER ULTRA SI | Cursor IDE | 47 |  Queue |

| ****DYN-009**** | DEXTER NLP artifact parser pipeline: raw document to structured artifact node – title, abstract extraction, key finding synthesis, domain tag assignment (D1-D8) | DEXTER ULTRA SI | claude-sonnet-4-6 | 50 |  Queue |

| ****DYN-010**** | Component relevance mapper: link artifact findings to BLEND library component IDs (part_id). Output: affected_component_ids[] with confidence score per mapping | DEXTER ULTRA SI | claude-sonnet-4-6 | 50 |  Queue |

| ****DYN-011**** | Regulatory impact extractor: identify specific regulation references, affected vehicle categories, compliance deadline dates, type approval pathway implications | DEXTER ULTRA SI | claude-sonnet-4-6 | 49 |  Queue |

| ****DYN-012**** | Novelty scorer: cosine similarity between new artifact embedding and existing knowledge graph. Score 0–1.0. Above 0.8 = high novelty, below 0.2 = known territory | DEXTER ULTRA SI | claude-sonnet-4-6 | 49 |  Queue |

| ****DYN-013**** | KOPA analytics relevance scorer: domain_relevance (0–1.0), component_library_impact (count), programme_urgency (deadline proximity), novelty_index, confidence_level | DEXTER ULTRA SI | claude-sonnet-4-6 | 50 |  Queue |

| ****DYN-014**** | Frontier zone detector: rolling 90-day discovery density baseline per domain. Flag FRONTIER_ZONE node when new discovery density exceeds baseline by >2-sigma | DEXTER ULTRA SI | claude-sonnet-4-6 | 50 |  Queue |

| ****DYN-015**** | PIECE MAKER frontier alert: when frontier zone intersects active BLEND gap (domain match + severity HIGH/CRITICAL), emit piecemaker.frontier_alert event to Redis Streams | DEXTER ULTRA SI | Cursor IDE | 49 |  Queue |

| ****DYN-016**** | DYNAMO Knowledge API (GraphQL): discoveries by domain query, component impact analysis query, frontier zone map endpoint, regulatory change timeline, research trend velocity | DEXTER ULTRA SI | Cursor IDE | 48 |  Queue |

| ****DYN-017**** | Push notification service (notification-svc, port 3080): WebSocket real-time push for DYNAMO discoveries. Priority routing: CRITICAL immediate, HIGH <1min, MEDIUM batch 5min | ADA ULTRA SI | Cursor IDE | 48 |  Queue |

| ****DYN-018**** | Discovery push message format: engine, priority, affected_components[], discovery_title, domain, frontier_zone_flag, call_to_action link to BLEND assembly view | ADA ULTRA SI | claude-sonnet-4-6 | 47 |  Queue |

| ****DYN-019**** | DYNAMO subscription service: POST /api/dynamo/subscribe — register session for real-time updates on specified domains. Persist subscription with TTL = session duration | ADA ULTRA SI | Cursor IDE | 47 |  Queue |

| ****DYN-020**** | DYNAMO research cycle metrics: write cycle performance to TimescaleDB research_pulses (documents_scanned, discoveries_found, graph_updates, frontier_detections) | ADA ULTRA SI | Firebase Studio | 46 |  Queue |

| ****DYN-021**** | Claude embeddings API integration: vectorize high-relevance artifacts (score > 0.7), store 1536-dim embeddings in Pinecone research_embeddings collection | DEXTER ULTRA SI | Cursor IDE | 48 |  Queue |

| ****DYN-022**** | Semantic similarity search endpoint: /api/dynamo/graph/similar — given a component spec or design brief, find related research artifacts via Pinecone cosine search | DEXTER ULTRA SI | Cursor IDE | 48 |  Queue |

| ****DYN-023**** | DYNAMO BLEND notification pipeline: when component library changes (new/updated/deprecated), emit blend.library_update event, trigger BLEND re-score for active assemblies | ADA ULTRA SI | Cursor IDE | 48 |  Queue |

| ****DYN-024**** | MongoDB artifact archive: store full document JSON (paper, patent, regulation) in dynamo_artifacts collection after graph integration. Index by domain, date, relevance_score | ADA ULTRA SI | Cursor IDE | 46 |  Queue |

| ****DYN-025**** | GRO DURC screening integration in DYNAMO: ANT KING evaluates each manufacturing process artifact for dual-use potential. KILLZONE trigger quarantines artifact from graph | ANT KING | claude-sonnet-4-6 | 50 |  Queue |

| ****DYN-026**** | Safety recall detection: DYNAMO regulatory crawler identifies recall-linked component IDs. SAFE_LIFE trigger flags components in BLEND library, user notification dispatched | ANT KING | claude-sonnet-4-6 | 50 |  Queue |

| ****DYN-027**** | DYNAMO dashboard (analytics-svc): real-time metrics — discovery rate by domain, frontier zone map, compliance deadline timeline, BLEND library freshness score | ADA ULTRA SI | Firebase Studio | 46 |  Queue |

| ****DYN-028**** | DYNAMO Grafana dashboard: research_pulses time-series visualisation, crawl worker queue depths, p95 parse time per source, frontier zone emergence heatmap | ADA ULTRA SI | Firebase Studio | 45 |  Queue |

| ****DYN-029**** | DYNAMO autoscaling: CPU-intensive NLP workers scale independently from API workers. HPA config: scale out when NLP queue > 50 jobs pending, scale in after 10min idle | ADA ULTRA SI | Cursor IDE | 46 |  Queue |

| ****DYN-030**** | DYNAMO error handling and retry: failed crawl retried 3× with exponential backoff. Failed parse artifacts staged in MongoDB error_artifacts for manual review queue | ADA ULTRA SI | Cursor IDE | 46 |  Queue |

| ****DYN-031**** | Ad-hoc research trigger: POST /api/dynamo/trigger — immediate research cycle for specified domain and artifact type. Priority queue insertion, result within 15 minutes | ADA ULTRA SI | Cursor IDE | 47 |  Queue |

| ****DYN-032**** | DYNAMO rate limiting and cost control: Claude API spend cap per research cycle (\$50/day), fallback to cached results if spend cap hit, alert to admin channel | ADA ULTRA SI | Cursor IDE | 46 |  Queue |

Phase 2B – Research Knowledge Graph (18 Prompts)

| Code | Prompt Specification | SPC Owner | VIBE Tool | JCSE | Status |

|---|---|---|---|---|---|

| **GRP-001** | Neo4j 5.17 database deployment: component graph schema migration – Component, Research, RegulationUpdate, FrontierZone, SupplierBulletin node types with property schemas | DEXTER ULTRA SI | Cursor IDE | 50 |  Queue |

| **GRP-002** | COMPATIBLE_WITH edge schema: part_id_a, part_id_b, synergy_score, test_validated (bool), weight, cost_delta, last_updated. Bi-directional relationship | ADA ULTRA SI | Cursor IDE | 49 |  Queue |

| **GRP-003** | SUPERSEDES edge schema: old_part_id, new_part_id, delta_weight_kg, delta_cost_usd, delta_co2e_kg, reason, effective_date. DYNAMO creates on spec update detection | ADA ULTRA SI | Cursor IDE | 49 |  Queue |

| **GRP-004** | SUPPORTED_BY edge schema: component_id to research_artifact_id – relevance_score, domain, finding_type (supports/challenges/extends), created_at timestamp | DEXTER ULTRA SI | Cursor IDE | 49 |  Queue |

| **GRP-005** | REGULATORY_IMPACT edge schema: regulation_id to component_id[] – market (EU/US/CN/IN/ROW), compliance_deadline, impact_type (mandatory/recommended), severity | DEXTER ULTRA SI | Cursor IDE | 49 |  Queue |

| **GRP-006** | REQUIRES_GAP edge schema: assembly_id to gap_specification – domain, severity (CRITICAL/HIGH/MEDIUM), spec_requirements, piece_maker_triggered (bool), frontier_zone_ref | ADA ULTRA SI | Cursor IDE | 49 |  Queue |

| **GRP-007** | FrontierZone node schema: zone_id, domain (D1-D8), discovery_count, rolling_baseline, sigma_excess, earliest_discovery, latest_discovery, active_piece_maker_gaps | DEXTER ULTRA SI | Cursor IDE | 50 |  Queue |

| **GRP-008** | Graph write service: Neo4j writer module that creates/updates nodes and edges from DYNAMO artifact parser output. Idempotent writes via MERGE on unique IDs | ADA ULTRA SI | Cursor IDE | 48 |  Queue |

| **GRP-009** | Graph seed data pipeline: initial 200,000 component library ingestion from automotive PDM exports. Batch Cypher LOAD CSV pipeline with progress tracking | ADA ULTRA SI | Cursor IDE | 47 |  Queue |

| **GRP-010** | BOM traversal algorithm: A* pathfinding from vehicle_spec node through 10 system categories (SYS-01 to SYS-10), weighted by synergy_score × objective_weight | ADA ULTRA SI | Cursor IDE | 50 |  Queue |

| **GRP-011** | Graph query optimisation: index on Component(part_id, category, vehicle_class_compat, powertrain_compat), text index on Component(name), composite index for BOM queries | ADA ULTRA SI | Cursor IDE | 47 |  Queue |

| **GRP-012** | DYNAMO graph freshness score: per-component metric tracking days since last DYNAMO update. BLEND BOM generator prioritises components with freshness_score < 30 days | DEXTER ULTRA SI | Cursor IDE | 47 |  Queue |

| ****GRP-013**** | Frontier detection algorithm: rolling window analytics query in Neo4j — count Research nodes per domain per 30-day window, compute 90-day baseline, flag >2-sigma outliers | DEXTER ULTRA SI | claude-sonnet-4-6 | 50 |  Queue |

| ****GRP-014**** | Frontier zone intersection check: cross-query FrontierZone nodes against REQUIRES_GAP edges — match on domain and severity. Output: priority PIECE MAKER job queue events | DEXTER ULTRA SI | claude-sonnet-4-6 | 50 |  Queue |

| ****GRP-015**** | DEXTER IRB methodology integration: research quality gating — peer_review_status, citation_count, journal_impact_factor minimum thresholds before graph integration | DEXTER ULTRA SI | claude-sonnet-4-6 | 49 |  Queue |

| ****GRP-016**** | DURC screening Neo4j integration: quarantine node type for flagged artifacts. GRO KILLZONE adds artifact to quarantine partition, logs ANT KING decision, excludes from all graph queries | ANT KING | claude-sonnet-4-6 | 50 |  Queue |

| ****GRP-017**** | Knowledge graph backup and disaster recovery: daily Neo4j dump to S3, point-in-time recovery to within 1 hour, cross-region replication for programme/enterprise tiers | ADA ULTRA SI | Cursor IDE | 46 |  Queue |

| ****GRP-018**** | Graph analytics dashboard: research coverage heatmap by BAHN domain, frontier zone velocity over time, component library age distribution, DYNAMO update frequency | ADA ULTRA SI | Firebase Studio | 45 |  Queue |

Phase 3 — BLEND Component Assembly (55 Prompts | Sprint 3 | Weeks 10–14 | JCSE Gate: 46+)

> Phase 3 builds BLEND — the Component Synergy Engine that maintains the 200,000+ automotive component library and assembles optimal vehicle Bills of Materials. BAHN × HOLMES ULTRA SI intelligence scores compatibility across 7 dimensions.

Phase 3A — BLEND Component Library & BOM Engine (35 Prompts)

| Code | Prompt Specification | SPC Owner | VIBE Tool | JCSE | Status |

|---|---|---|---|---|---|

| ****BLD-001**** | Deploy blend-svc (port 3020): NestJS module with component library query engine, BOM generator job queue, synergy scoring worker pool, regulatory compliance cache | BAHN ULTRA SI | Cursor IDE | 50 |  Queue |

| ****BLD-002**** | Component library ingestion pipeline: batch importer for 200,000+ automotive component nodes. CSV/JSON input format, validation against schema, Neo4j MERGE upsert | ADA ULTRA SI | Cursor IDE | 48 |  Queue |

| ****BLD-003**** | Component node schema validator: enforce required fields (part_id, name, category, supplier, mass_kg, co2e_kg, regulatory_status map, powertrain_compat[], vehicle_class_compat[]) | ADA ULTRA SI | Cursor IDE | 47 |  Queue |

| ****BLD-004**** | Component library search API: GET /api/blend/components — filter by domain, vehicle class (VC-A through VC-J), powertrain, supplier, year, category. Full-text + semantic search | ADA ULTRA SI | Cursor IDE | 47 |  Queue |

| ****BLD-005**** | Pinecone component embedding pipeline: vectorize component spec_text (name + category + properties + supplier description) and store in component_embeddings collection | ADA ULTRA SI | Cursor IDE | 48 |  Queue |

| ****BLD-006**** | State-of-the-art (SOTA) component sub-library: dedicated category for DYNAMO-sourced research-stage and prototype components. SOTA flag in component node, freshness tracking | DEXTER ULTRA SI | claude-sonnet-4-6 | 49 |  Queue |

| ****BLD-007**** | Component freshness management: DYNAMO update trigger refreshes component node properties, updates dynamo_last_updated timestamp, triggers re-score for active assemblies using this part | DEXTER ULTRA SI | Cursor IDE | 48 |  Queue |

| ****BLD-008**** | SYS-01 Body-In-White (BIW) category: structure nodes, crashworthiness ratings, material specifications, joining process compatibility, mass per zone targets | BAHN ULTRA SI | claude-sonnet-4-6 | 49 |  Queue |

| ****BLD-009**** | SYS-02 through SYS-05 categories: Chassis/Suspension, Powertrain/Drivetrain, Electrical/Electronics, Thermal Management — with system-specific property schemas | BAHN ULTRA SI | claude-sonnet-4-6 | 49 |  Queue |

| ****BLD-010**** | SYS-06 through SYS-10 categories: Interior/HMI, Exterior/Glazing, Software/ADAS, Safety Systems, Sealing/NVH — with domain-specific compatibility rules and regulatory flags | BAHN ULTRA SI | claude-sonnet-4-6 | 49 |  Queue |

| ****BLD-011**** | BOM generator core: POST /api/blend/assemble — accept vehicle spec (class, powertrain, markets, sustainability targets, budget), trigger BAHN domain routing, launch Neo4j traversal | ADA ULTRA SI | Cursor IDE | 50 |  Queue |

| ****BLD-012**** | BAHN domain routing for BLEND: parse vehicle spec, activate domain combination (e.g., BEV compact EU/US routes D1+D2+D4+D5+D6+D8), export domain constraints to BOM query | BAHN ULTRA SI | claude-sonnet-4-6 | 50 |  Queue |

| ****BLD-013**** | A* BOM traversal: Neo4j query from vehicle_spec node through 10 SYS categories. Weighted by: synergy_score × (objective_weight: 0.4 weight / 0.3 cost / 0.3 CO2e balanced) | ADA ULTRA SI | Cursor IDE | 50 |  Queue |

| ****BLD-014**** | DYNAMO-fresh component prioritisation: BOM traversal query adds freshness_score weight — prefer components with DYNAMO update within 30 days over stale library entries | DEXTER ULTRA SI | Cursor IDE | 48 |  Queue |

| ****BLD-015**** | SOTA component injection: when SOTA components exist for a system category meeting spec constraints, include as alternative BOM option alongside production-stage selections | DEXTER ULTRA SI | claude-sonnet-4-6 | 49 |  Queue |

| ****BLD-016**** | Synergy scoring engine: BAHN × HOLMES ULTRA SI scoring for each component pair — mechanical interface (25%), thermal (15%), electrical (15%), mass-stiffness (20%), NVH (10%), assembly (10%), regulatory (5%) | BAHN ULTRA SI | claude-sonnet-4-6 | 50 |  Queue |

| ****BLD-017**** | HOLMES deductive analysis layer: root cause diagnosis when synergy score < 65. Identify incompatibility type, generate plain-language explanation, suggest alternative components | BAHN ULTRA SI |

claude-sonnet-4-6 | 50 |  Queue |

| ****BLD-018**** | BOM Synergy Score aggregation: compute overall BOM Synergy Score (0-100) as weighted average of all component pair scores. Score < 65 triggers HOLMES re-optimisation | BAHN ULTRA SI | claude-sonnet-4-6 | 49 |  Queue |

| ****BLD-019**** | Component pair synergy cache: cache synergy scores in Redis for component pairs tested in the last 24 hours. Cache invalidated when either component receives DYNAMO update | ADA ULTRA SI | Cursor IDE | 46 |  Queue |

| ****BLD-020**** | Synergy score explanation generator: for each sub-optimal pair score (<70), produce human-readable compatibility analysis for the engineering team review interface | BAHN ULTRA SI | claude-sonnet-4-6 | 48 |  Queue |

| ****BLD-021**** | Regulatory compliance checker: for each assembled component, cross-reference regulatory_status map against target markets in session spec. Flag non-compliant components immediately | BAHN ULTRA SI | claude-sonnet-4-6 | 50 |  Queue |

| ****BLD-022**** | Multi-market compliance matrix: EU type approval (EU 2018/858), FMVSS (US), GB standards (China), CMVR (India), matrix output with market-by-market status per component | BAHN ULTRA SI | claude-sonnet-4-6 | 50 |  Queue |

| ****BLD-023**** | Regulatory deadline tracker: highlight components with compliance changes in next 90 days (HIGH urgency), 180 days (MEDIUM). Link to DYNAMO regulatory discovery | BAHN ULTRA SI | claude-sonnet-4-6 | 49 |  Queue |

| ****BLD-024**** | REACH substance check per component: cross-reference component material composition against ECHA REACH SVHC list. Flag substances of very high concern in BOM report | BAHN ULTRA SI | Cursor IDE | 49 |  Queue |

| ****BLD-025**** | Gap detection engine: after BOM traversal, identify system positions where no library component meets all constraints (spec + synergy_score > 70 + regulatory_status) | ADA ULTRA SI | Cursor IDE | 50 |  Queue |

| ****BLD-026**** | GAP_NODE creation in Neo4j: REQUIRES_GAP edge with gap specification (required_function, load_case, mass_target, cost_ceiling, CO2e_budget, regulatory_requirements, domain) | ADA ULTRA SI | Cursor IDE | 50 |  Queue |

| ****BLD-027**** | Gap severity classifier: CRITICAL (primary structure, safety system, high-voltage architecture), HIGH (powertrain, thermal management), MEDIUM (interior, NVH, sealing) | BAHN ULTRA SI | claude-sonnet-4-6 | 50 |  Queue |

| ****BLD-028**** | PIECE MAKER trigger event: emit piecemaker.gap_detected to Redis Streams with full gap specification. Include frontier_zone_ref if DYNAMO has relevant research for this gap domain | ADA ULTRA SI | Cursor IDE | 49 |  Queue |

| ****BLD-029**** | CO2e scoring integration: per-component CO2e_kg from material composition + manufacturing process. BOM total lifecycle CO2e/km calculation using BAHN MCS + BAHN D8 LCA | BAHN ULTRA SI | claude-sonnet-4-6 | 49 |  Queue |

| ****BLD-030**** | Assembly optimiser — LIGHTWEIGHT objective: substitute components to minimise total unsprung + sprung mass while maintaining synergy_score > 65 and regulatory compliance | BAHN ULTRA SI | claude-sonnet-4-6 | 49 |  Queue |

| ****BLD-031**** | Assembly optimiser — ECO objective: minimise total lifetime CO2e/km via MCS-derived substitutions. Priority: recycled/bio-based materials, low-energy manufacturing processes | BAHN ULTRA SI | claude-sonnet-4-6 | 49 |  Queue |

| ****BLD-032**** | Assembly optimiser — VALUE objective: minimise total programme cost (component cost + manufacturing + logistics) at target programme volume. Break-even analysis vs. SOTA alternatives | BAHN ULTRA SI | claude-sonnet-4-6 | 49 |  Queue |

| ****BLD-033**** | Optimiser delta report: after each optimisation pass, output which components changed, mass/cost/CO2e impact per change, overall BOM metric improvement summary | BAHN ULTRA SI | claude-sonnet-4-6 | 48 |  Queue |

| ****BLD-034**** | LCA mandatory enforcement: every BLEND assembly must complete BAHN D8 LCA (ISO 14040/44) before certification. LCA incomplete = BOM cannot proceed to FORGE | BAHN ULTRA SI | claude-sonnet-4-6 | 50 |  Queue |

| ****BLD-035**** | BLEND assembly GRO mode: CONTAINMENT trigger when BIW structure fails Euro NCAP minimum, or when high-voltage architecture missing isolation. Blocks assembly completion | ANT KING | claude-sonnet-4-6 | 50 |  Queue |

Phase 3B — BLEND Assembly Optimisation & Export (20 Prompts)

| Code | Prompt Specification | SPC Owner | VIBE Tool | JCSE | Status |

|---|---|---|---|---|---|

| ****ASM-001**** | BOM export — JSON format: complete bill of materials with component tree, synergy matrix, gap_report, regulatory_status per market, CO2e_summary, frontier_zone_refs | ADA ULTRA SI | Cursor IDE | 47 |  Queue |

| ****ASM-002**** | BOM export — CSV format: flat component list with key metrics (part_id, name, supplier, mass_kg, cost_usd, co2e_kg, synergy_score, regulatory_status, dynamo_last_updated) | ADA ULTRA SI | Cursor IDE | 46 |  Queue |

| ****ASM-003**** | BOM export — PDF format: formatted engineering document with synergy matrix visualisation, gap highlight, regulatory compliance summary, MCS sustainability score, FORGE watermark | SKRIBE SPC | Firebase Studio | 47 |  Queue |

| ****ASM-004**** | BOM versioning system: diff tracking between BOM versions (v1, v2, v3). Track which components changed, delta metrics (mass, cost, CO2e), reason codes for each change | ADA ULTRA SI | Cursor IDE | 47 |  Queue |

| ****ASM-005**** | BOM snapshot storage: approved BOM snapshots stored in MongoDB blend_bom_snapshots with full component tree, synergy matrix, and immutable hash for IP provenance | ADA ULTRA SI | Cursor IDE | 47 |  Queue |

| ****ASM-006**** | PLM integration endpoint — Teamcenter: REST adapter for BLEND BOM export to Siemens Teamcenter API. Component mapping to PLM part numbering schema | ADA ULTRA SI | Cursor IDE | 46 |  Queue |

| ****ASM-007**** | PLM integration endpoint — Windchill: REST adapter for PTC Windchill API. BOM structure mapping, change management integration, ECO (Engineering Change Order) triggers | ADA ULTRA SI | Cursor IDE | 46 |  Queue |

| ****ASM-008**** | PLM integration endpoint — Enovia: SOAP/REST adapter for Dassault Systèmes 3DEXPERIENCE. BOM federation, component attribute mapping, lifecycle state synchronisation | ADA ULTRA SI | Cursor IDE | 46 |  Queue |

| ****ASM-009**** | FORGE certification trigger for BOM: on BOM completion with synergy_score > 80, submit to forge-svc for JCSE scoring, HIVE 14-D audit, certification generation | ADA ULTRA SI | Cursor IDE | 48 |  Queue |

| ****ASM-010**** | NFT snapshot certification: certified BOM hash submitted to nft-svc for ERC-721 mint on Polygon L2. Token metadata: vehicle_class, powertrain, synergy_score, domains, timestamp | ADA ULTRA SI | Cursor IDE | 47 |  Queue |

| ****ASM-011**** | DYNAMO subscription for assembly: after BOM completion, auto-subscribe the session to relevant domain updates (based on BOM component domains). Live until session closes | DEXTER ULTRA SI | Cursor IDE | 47 |  Queue |

| ****ASM-012**** | BLEND session state management: persist partial assembly in Redis session_cache. Resume assembly from last checkpoint after connection interruption or browser refresh | ADA ULTRA SI | Cursor IDE | 46 |  Queue |

| ****ASM-013**** | BLEND performance targets: BOM generation < 30 seconds for 200-component vehicle. Synergy scoring < 10 seconds per assembly. Gap detection < 5 seconds after traversal | ADA ULTRA SI | Firebase Studio | 46 |  Queue |

| ****ASM-014**** | BLEND GRO weaponisation guard: if session spec indicates hostile deployment or combat platform, GRO KILLZONE terminates session immediately. Zero BOM output delivered | ANT KING | claude-sonnet-4-6 | 50 |  Queue |

| ****ASM-015**** | GET /api/blend/assembly/{id}/bom: retrieve full BOM with synergy matrix and component compatibility graph. Response includes gap_count, synergy_score, certification_status | ADA ULTRA SI | Cursor IDE | 46 |  Queue |

| ****ASM-016**** | GET /api/blend/assembly/{id}/gaps: list all component gaps with severity (CRITICAL/HIGH/MEDIUM), domain, required specification, frontier_zone_ref, PIECE MAKER job status | ADA ULTRA SI | Cursor IDE | 46 |  Queue |

| ****ASM-017**** | POST /api/blend/optimise: run optimiser on existing assembly with specified objective (LIGHTWEIGHT/ECO/VALUE). Return delta report with before/after metric comparison | ADA ULTRA SI | Cursor IDE | 47 |  Queue |

| ****ASM-018**** | BLEND analytics: design_events TimescaleDB writes for assembly start, domain routing, synergy scores, gap detections, optimiser runs. Feed analytics-svc dashboard | ADA ULTRA SI | Firebase Studio | 45 |  Queue |

| ****ASM-019**** | BLEND accuracy metrics: track synergy_score prediction accuracy vs. engineer-validated assemblies. KOPA analytics module measures BLEND improvement over time | DEXTER ULTRA SI | Firebase Studio | 46 |  Queue |

| ****ASM-020**** | BLEND PIECE MAKER integration confirmation: after PIECE MAKER closes a gap and adds new component node to Neo4j, BLEND re-scores the assembly with the new component | ADA ULTRA SI | Cursor IDE | 48 |  Queue |

Phase 4 – PIECE MAKER Gap Manufacturing (58 Prompts | Sprint 4 | Weeks 15–19 | JCSE Gate: 48+)

> Phase 4 builds PIECE MAKER – the generative design-to-manufacture pipeline. PIECE MAKER does not identify gaps: it resolves them. DEXTER ULTRA SI injects DYNAMO frontier research into every design brief. BAHN × FRACTAL ULTRA SI power topology optimisation and material selection.

Phase 4A – PIECE MAKER Core Pipeline (30 Prompts)

Code	Prompt Specification	SPC Owner	VIBE Tool	JCSE	Status
	--- --- --- --- ---				
PM-001	Deploy piecemaker-svc (port 3030): NestJS module with gap event consumer (Redis Streams), job orchestrator, BullMQ queues for generative design, material selection, compliance	ADA ULTRA SI	Cursor IDE	50	Queue
PM-002	Gap receiver module: consume piecemaker.gap_detected events from Redis Streams. Parse full gap specification, create PIECEMAKER_JOB record in PostgreSQL with status=RECEIVED	ADA ULTRA SI	Cursor IDE	50	Queue
PM-003	Gap severity escalation: CRITICAL gaps (primary structure, safety, HV architecture) immediately notify Senior Engineer and Programme Manager via notification-svc. GRO SAFE_LIFE activated	ANT KING	Cursor IDE	50	Queue
PM-004	Gap classification by BAHN domain (D1-D8): route gap job to appropriate BAHN mode for engineering constraint definition. D2 gaps route to BAHN:ENGINEER, D5 to BAHN:MATERIALS	BAHN ULTRA SI	claude-sonnet-4-6	50	Queue
PM-005	GET /api/piecemaker/job/{id}/status: poll PIECE MAKER job progress. Return current step (1-7), completion percentage, time elapsed, estimated completion, current action	ADA ULTRA SI	Cursor IDE	48	Queue
PM-006	PIECE MAKER job queue management: priority ordering – CRITICAL gap jobs before HIGH before MEDIUM. Max 5 concurrent CRITICAL jobs. Queue depth monitoring and SLA alerts	ADA ULTRA SI	Cursor IDE	47	Queue
PM-007	DYNAMO frontier intelligence query: before generative design, query Neo4j for FrontierZone nodes matching gap domain. If found, retrieve top-5 research artifacts as design brief inputs	DEXTER ULTRA SI	claude-sonnet-4-6	50	Queue
PM-008	Research-to-design-brief translator: DEXTER converts frontier research findings into engineering design brief inputs – candidate materials, process hints, reference geometry parameters	DEXTER ULTRA SI	claude-sonnet-4-6	50	Queue
PM-009	Frontier injection confirmation: if DYNAMO frontier research is injected into design brief, tag PIECEMAKER_JOB with frontier_informed=true and list artifact_ids used	DEXTER ULTRA SI	Cursor IDE	48	Queue
PM-010	No-frontier fallback: when no DYNAMO frontier zone exists for gap domain, PIECE MAKER proceeds with BAHN engineering constraints only. Standard design brief without research injection	ADA ULTRA SI	Cursor IDE	47	Queue

| ****PM-011**** | Frontier alert webhook: when new DYNAMO frontier zone emerges in a domain with open
PIECE MAKER gaps, emit alert to Session and PM channel for potential design brief enrichment | DEXTER
ULTRA SI | Cursor IDE | 47 |  Queue |

| ****PM-012**** | Frontier zone research quality filter: apply DEXTER IRB quality gates to frontier research before
injection — min citation count 10, peer-reviewed only, published < 24 months | DEXTER ULTRA SI | claude-
sonnet-4-6 | 49 |  Queue |

| ****PM-013**** | Design brief formulator: DEXTER generates structured engineering brief —
functional_requirements, load_case, interface_geometry, mass_target, cost_ceiling, co2e_budget,
regulatory_checklist | DEXTER ULTRA SI | claude-sonnet-4-6 | 50 |  Queue |

| ****PM-014**** | Load path requirements: BAHN:ENGINEER provides load case inputs for structural
components — peak load (N), fatigue cycles, safety factor, failure mode constraints (no fracture before
200kN) | BAHN ULTRA SI | claude-sonnet-4-6 | 50 |  Queue |

| ****PM-015**** | Interface geometry specification: extract interface constraints from BLEND assembly — mating
component part_id, interface type (bolt pattern, weld flange, press fit), tolerance stack | ADA ULTRA SI |
claude-sonnet-4-6 | 49 |  Queue |

| ****PM-016**** | DfAM guidelines injection: BAHN D3 manufacturing domain provides Design-for-Additive-
Manufacture rules — max overhang angle (45° for FDM), min wall thickness (0.8mm for DMLS), build
orientation | BAHN ULTRA SI | claude-sonnet-4-6 | 49 |  Queue |

| ****PM-017**** | Design brief schema: structured JSON output from formulator — required fields validated
before topology optimisation launch. Schema includes all functional, load, interface, material, process fields |
ADA ULTRA SI | Cursor IDE | 48 |  Queue |

| ****PM-018**** | ADA data architecture for design brief: define MongoDB piecemaker_designs schema.
Versioned brief storage with diff tracking as brief is enriched through Steps 2-3 | ADA ULTRA SI | Cursor IDE |
47 |  Queue |

| ****PM-019**** | Topology optimisation engine: FRACTAL ULTRA SI Sierpinski lattice generator — apply load
path inputs to multi-objective topology optimisation (mass vs. stiffness vs. cost vs. CO2e) | FRACTAL ULTRA
SI | claude-sonnet-4-6 | 50 |  Queue |

| ****PM-020**** | DfAM constraint filter: post-topology geometry validation against selected AM process
constraints. Reject geometries with unsupported overhangs >45° (FDM), walls <0.8mm (DMLS) | FRACTAL
ULTRA SI | claude-sonnet-4-6 | 50 |  Queue |

| ****PM-021**** | Generative design candidate ranking: score topology candidates on combined
mass_reduction%, stiffness_retention%, cost_estimate, co2e_estimate, AM_feasibility. Top 3 candidates
returned | FRACTAL ULTRA SI | claude-sonnet-4-6 | 49 |  Queue |

| ****PM-022**** | Sierpinski lattice structure generation: internal geometry optimisation for additive
manufactured components — progressive void distribution for mass reduction preserving load paths |
FRACTAL ULTRA SI | claude-sonnet-4-6 | 49 |  Queue |

| ****PM-023**** | CAD reference generation: produce parametric geometry specification (not full CAD file) —
dimensions, topology description, key cross-sections, interface geometry confirmation | FRACTAL ULTRA SI |
claude-sonnet-4-6 | 48 |  Queue |

| ****PM-024**** | FRACTAL factory cell design: for components entering volume production, FRACTAL generates
Industry 4.0 factory cell layout — AM machine placement, post-processing station, quality inspection |

FRACTAL ULTRA SI | claude-sonnet-4-6 | 49 |  Queue |

| ****PM-025**** | Material shortlist generator: BAHN MCS tool produces candidate material list meeting design brief – structural performance (yield, fatigue, thermal), CO2e/kg budget, REACH compliance | BAHN ULTRA SI | claude-sonnet-4-6 | 50 |  Queue |

| ****PM-026**** | AM process selector (AMS): BAHN AMS decision matrix – DMLS for structural metal (Ti-6Al-4V, AlSi10Mg), FDM/SLA for polymers (PEEK, PA12), SLS for medium-complexity polymer | BAHN ULTRA SI | claude-sonnet-4-6 | 50 |  Queue |

| ****PM-027**** | Material × process compatibility validator: ensure selected material is compatible with recommended AM process – DMLS alloys list, FDM thermoplastic compatibility, SLS powder flowability | BAHN ULTRA SI | claude-sonnet-4-6 | 49 |  Queue |

| ****PM-028**** | Supplier availability check: query DYNAMO knowledge graph for current supplier availability of shortlisted materials – ExOne, EOS, Stratasys supplier nodes with current spec and lead time | DEXTER ULTRA SI | claude-sonnet-4-6 | 48 |  Queue |

| ****PM-029**** | Material Specification Sheet output: BAHN MCS score, CO2e/kg, supplier, AM compatibility, yield strength, fatigue limit, thermal range, REACH status, sourcing lead time | BAHN ULTRA SI | claude-sonnet-4-6 | 49 |  Queue |

| ****PM-030**** | AM Bill of Process output: selected AM process, machine parameters (layer thickness, scan speed, energy density), support strategy, post-processing steps (HIP, machining, NDT requirements) | BAHN ULTRA SI | claude-sonnet-4-6 | 49 |  Queue |

Phase 4B – PIECE MAKER Integration, Certification & Graph Update (28 Prompts)

| Code | Prompt Specification | SPC Owner | VIBE Tool | JCSE | Status |

|---|---|---|---|---|---|

| ****PMI-001**** | Business case calculator: cost-per-part at 4 programme volumes (1K/5K/10K/50K units/year) using AM process cost models. Compare to conventional process (cast/stamped/forged) | BAHN ULTRA SI | claude-sonnet-4-6 | 49 |  Queue |

| ****PMI-002**** | Tooling avoidance analysis: calculate tooling cost avoided if PIECE MAKER design replaces a cast or stamped part. Amortise over 5-year programme at target volume | BAHN ULTRA SI | claude-sonnet-4-6 | 48 |  Queue |

| ****PMI-003**** | Break-even volume calculation: at which annual volume does conventional manufacture become cheaper than AM? Output crossover point and recommendation | BAHN ULTRA SI | claude-sonnet-4-6 | 48 |  Queue |

| ****PMI-004**** | Weight saving vehicle-level impact: PIECE MAKER mass delta propagated to vehicle CO2e/km calculation. Output: gCO2e/km saved per kg reduced at WLTP cycle | BAHN ULTRA SI | claude-sonnet-4-6 | 49 |  Queue |

| ****PMI-005**** | Lead time vs. conventional benchmark: AM lead time (prototype: 1–4 weeks, production ramp: 8–16 weeks) vs. conventional tooling + production lead time per process | BAHN ULTRA SI | claude-sonnet-4-6 | 47 |  Queue |

| ****PMI-006**** | Business Case Summary document: formatted JSON/PDF – production economics table, break-even chart data, weight-saving CO2e table, programme recommendation, DYNAMO link | SKRIBE SPC |

Firebase Studio | 47 |  Queue |

| ****PMI-007**** | Compliance validator: BAHN Phase 6 homologation intelligence applied to PIECE MAKER component – does material + process combination meet regulatory requirements for target markets? | BAHN ULTRA SI | claude-sonnet-4-6 | 50 |  Queue |

| ****PMI-008**** | REACH substance validation: cross-reference AM material chemical composition against ECHA REACH SVHC candidate list and restricted substances list. Flag any hits, propose alternatives | BAHN ULTRA SI | claude-sonnet-4-6 | 50 |  Queue |

| ****PMI-009**** | Type approval implication report: for PIECE MAKER component replacing an existing certified component, identify type approval re-submission requirements per market (EU/US/CN/IN) | BAHN ULTRA SI | claude-sonnet-4-6 | 49 |  Queue |

| ****PMI-010**** | GRO HUMAN_IN_LOOP for CRITICAL components: design brief paused after Step 3 (brief generation). Dispatch engineer notification with brief summary. Resume only after sign-off | ANT KING | Cursor IDE | 50 |  Queue |

| ****PMI-011**** | GRO SAFE_LIFE mode: for structural and safety-critical components, require NDT specification (CT scan, dye penetrant, ultrasonic) in AM Bill of Process before output package | ANT KING | claude-sonnet-4-6 | 50 |  Queue |

| ****PMI-012**** | GRO KILLZONE for PIECE MAKER: if gap description contains weaponisation indicators (armour-piercing, ballistic, military offensive), job terminated immediately. ARES protocol engaged | ANT KING | claude-sonnet-4-6 | 50 |  Queue |

| ****PMI-013**** | Regulatory Compliance Memo output: per-market compliance status, REACH substance report, type approval pathway, safety flags, engineer sign-off requirement (if CRITICAL) | BAHN ULTRA SI | claude-sonnet-4-6 | 49 |  Queue |

| ****PMI-014**** | PIECE MAKER Output Package assembler: compile all outputs into single deliverable – Generative Design Report, Material Spec Sheet, AM Bill of Process, Business Case, Compliance Memo | ADA ULTRA SI | Cursor IDE | 49 |  Queue |

| ****PMI-015**** | DYNAMO link injection: append to output package which DYNAMO research discoveries informed the design (frontier_artifact_ids[]). Enables design traceability to scientific sources | DEXTER ULTRA SI | Cursor IDE | 49 |  Queue |

| ****PMI-016**** | Output package storage: JSON in MongoDB piecemaker_designs, PDF in S3 with per-session encryption, CAD reference in S3 3D-assets bucket, business case in PostgreSQL piecemaker_jobs | ADA ULTRA SI | Cursor IDE | 47 |  Queue |

| ****PMI-017**** | FORGE certification trigger: POST /api/forge/certify with PIECE MAKER output package. forge-svc returns JCSE score, HIVE 14-D audit result, certification document | ADA ULTRA SI | Cursor IDE | 48 |  Queue |

| ****PMI-018**** | forge-svc (port 3060) JCSE scoring engine: evaluate PIECE MAKER output across 10 JCSE dimensions. Score 0-50. Platinum (46-50) required for PIECE MAKER CRITICAL components | ADA ULTRA SI | claude-sonnet-4-6 | 50 |  Queue |

| ****PMI-019**** | HIVE 14-D audit for PIECE MAKER output: evaluate across all 14 HIVE dimensions. Target: Accuracy 95+, Depth 98+, Ethical Compliance 100, Safety Architecture 100 | ADA ULTRA SI | claude-sonnet-4-6 | 49 |  Queue |

| ****PMI-020**** | NFT certification mint: nft-svc (port 3095) mints ERC-721 token on Polygon L2. Metadata: component_spec_hash, material, AM_process, JCSE_score, HIVE_score, design_date | ADA ULTRA SI | Cursor IDE | 47 |  Queue |

| ****PMI-021**** | IPFS metadata storage: certified component specification stored on IPFS. Token URI points to IPFS CID. Enables decentralised, immutable design provenance verification | ADA ULTRA SI | Cursor IDE | 46 |  Queue |

| ****PMI-022**** | PCODEX IP protection: PIECE MAKER generative designs encoded via PCODEX before external output. Stenographic compression + cryptographic encoding. IP provenance timestamp | ADA ULTRA SI | Cursor IDE | 47 |  Queue |

| ****PMI-023**** | DYNAMO re-ingestion: completed PIECE MAKER designs added to DYNAMO knowledge graph as DESIGNED_BY_PIECEMAKER nodes. Future DYNAMO crawls include this as manufactured knowledge | DEXTER ULTRA SI | Cursor IDE | 49 |  Queue |

| ****PMI-024**** | BLEND re-ingestion: new PIECE MAKER component added to Neo4j component library. BLEND notified via blend.component_added event. Available for future assemblies immediately | ADA ULTRA SI | Cursor IDE | 49 |  Queue |

| ****PMI-025**** | DYNAMO subscribe for designed component: post PIECE MAKER completion, subscribe new component to DYNAMO domain updates. Component freshness tracked from creation | DEXTER ULTRA SI | Cursor IDE | 47 |  Queue |

| ****PMI-026**** | GET /api/piecemaker/job/{id}/result: retrieve completed PIECE MAKER output package – all documents, certification references, NFT token address, FORGE certificate ID | ADA ULTRA SI | Cursor IDE | 47 |  Queue |

| ****PMI-027**** | PIECE MAKER analytics: job completion times by severity, material selection frequency, AM process distribution, DYNAMO frontier injection rate, FORGE certification pass rate | ADA ULTRA SI | Firebase Studio | 45 |  Queue |

| ****PMI-028**** | PIECE MAKER Grafana dashboard: active jobs by status, CRITICAL gap queue depth, JCSE score distribution, ANT KING intervention frequency, average job completion time | ADA ULTRA SI | Firebase Studio | 45 |  Queue |

Phase 5 – Governance, Deploy & Certify (27 Prompts | Sprints 5–6 | Weeks 20–28 | JCSE Gate: 50/50)

> Phase 5 completes the AUTODRAFT production system. Royal DNA Protocol integration-tested across all three engines. Full multi-region deployment validated. HIVE 14-D Platinum audit targets 97.6/100. FORGE JCSE 50/50 certification obtained. SPHINX marketplace listing activated.

GRO Royal DNA Governance (8 Prompts)

| Code | Prompt Specification | SPC Owner | VIBE Tool | JCSE | Status |

|---|---|---|---|---|---|

| ****GOV-001**** | Royal DNA Protocol integration test: execute KILLZONE scenario for each engine (DYNAMO DURC, BLEND weaponised vehicle, PIECE MAKER offensive component). Verify zero output delivered | ANT KING | claude-sonnet-4-6 | 50 |  Queue |

| ****GOV-002**** | GRO SAFE_LIFE integration test: trigger safety recall scenario in DYNAMO crawler. Verify affected components flagged in BLEND, user notification dispatched within 60 seconds | ANT KING | claude-sonnet-4-6 | 50 |  Queue |

| ****GOV-003**** | GRO CONTAINMENT integration test: trigger Euro NCAP crashworthiness failure in BLEND BIW assembly. Verify assembly blocked, BAHN-SE escalation dispatched, no BOM delivered | ANT KING | claude-sonnet-4-6 | 50 |  Queue |

| ****GOV-004**** | GRO HUMAN_IN_LOOP integration test: submit CRITICAL structural gap to PIECE MAKER. Verify design brief paused after Step 3, engineer notified, design only resumes after sign-off | ANT KING | claude-sonnet-4-6 | 50 |  Queue |

| ****GOV-005**** | ANT QUEEN v2 Baby Agent factory: configure DYNAMO, BLEND, and PIECE MAKER specialist sub-agent spawning at Newborn stage. Define 6-stage evolution milestones per sub-agent colony | ANT QUEEN v2 | claude-sonnet-4-6 | 49 |  Queue |

| ****GOV-006**** | Enterprise colony spawning: ANT QUEEN v2 departmental colony architecture – separate DYNAMO colonies per BAHN domain, BLEND colony per vehicle programme, PIECE MAKER colony per factory | ANT QUEEN v2 | claude-sonnet-4-6 | 49 |  Queue |

| ****GOV-007**** | GRO audit trail validation: verify all GRO state transitions (LIFE/SAFE_LIFE/CONTAINMENT/KILLZONE) are logged in PostgreSQL auth_svc audit table with full context snapshot | ANT KING | Cursor IDE | 50 |  Queue |

| ****GOV-008**** | EU AI Act conformity assessment: document AUTODRAFT as High-Risk AI system per Article 6. Validate human oversight records, technical documentation, accuracy metrics per Annex IV | STRATEGOS ULTRA SI | claude-sonnet-4-6 | 49 |  Queue |

Deployment & Infrastructure (8 Prompts)

| Code | Prompt Specification | SPC Owner | VIBE Tool | JCSE | Status |

|---|---|---|---|---|

| ****DEP-001**** | Multi-region deployment: AWS us-east-1 (primary) + eu-west-1 (EU data residency) + ap-southeast-1 (Asia). Active-active for enterprise tier, active-passive for programme tier | ADA ULTRA SI | Cursor IDE | 47 |  Queue |

| ****DEP-002**** | Kubernetes autoscaling validation: load test all 12 services at 500 concurrent sessions. Verify HPA scale-out triggers for dynamo-svc (CPU) and blend-svc (memory) | ADA ULTRA SI | Firebase Studio | 47 |  Queue |

| ****DEP-003**** | Microservices health endpoint audit: each service exposes /health, /ready, /metrics endpoints. Kong health check configured. Kubernetes liveness + readiness probes confirmed | ADA ULTRA SI | Cursor IDE | 46 |  Queue |

| ****DEP-004**** | End-to-end integration test suite: run full DYNAMO-BLEND-PIECE MAKER flywheel. DYNAMO discovery triggers BLEND update, BLEND assembly finds gap, PIECE MAKER closes gap, FORGE certifies | ADA ULTRA SI | Cursor IDE | 48 |  Queue |

| ****DEP-005**** | Performance SLA validation: BAHN CONCEPT <3s, BAHN SOVEREIGN <15s, BLEND assemble <30s, DYNAMO push <1min, PIECE MAKER CRITICAL job <4hrs, NFT mint <2min | ADA ULTRA SI | Firebase Studio | 46 |  Queue |

| ****DEP-006**** | Security penetration test checklist: JWT token hijack, RBAC privilege escalation, Neo4j injection, Redis queue poisoning, PIECE MAKER job spoofing, NFT contract audit | ADA ULTRA SI | Cursor IDE | 47 |  Queue |

| ****DEP-007**** | Design Studio tier launch validation: single-region EKS 3-node cluster, \$800–1,200/month cost confirmed, Auth working, BAHN:CONCEPT + BAHN:ENGINEER operational, first BLEND assembly successful | ADA ULTRA SI | Firebase Studio | 46 |  Queue |

| ****DEP-008**** | Programme Team tier launch: dual-region, autoscaling 6–12 nodes, TimescaleDB live, full observability stack, SAML SSO validated, GDPR EU residency enforced, 500 concurrent sessions | ADA ULTRA SI | Firebase Studio | 46 |  Queue |

HIVE 14-D Certification (6 Prompts)

| Code | Prompt Specification | SPC Owner | VIBE Tool | JCSE | Status |

|---|---|---|---|---|---|

| ****HIVE-001**** | HIVE 14-D Platinum audit — Dimensions 1–7 (ADA ULTRA SI): Accuracy (95), Depth (98), Breadth (97), Actionability (96), Ethical Compliance (100), Innovation (98), Safety Architecture (100) | ADA ULTRA SI | claude-sonnet-4-6 | 50 |  Queue |

| ****HIVE-002**** | HIVE 14-D Platinum audit — Dimensions 8–11 (DEXTER ULTRA SI): Research Quality (97), Sustainability (98), Regulatory Intelligence (95), Technical Rigour (97) | DEXTER ULTRA SI | claude-sonnet-4-6 | 50 |  Queue |

| ****HIVE-003**** | HIVE 14-D Platinum audit — Dimensions 12–14 (BAHN ULTRA SI): Communication Quality (96), Integration (97), Ecosystem Fit (98). Overall target: 97.6/100 Platinum | BAHN ULTRA SI | claude-sonnet-4-6 | 50 |  Queue |

| ****HIVE-004**** | FORGE pipeline certification: submit AUTODRAFT platform SDD + PDD + all Phase deliverables to JCSE scoring. Target JCSE 50/50 Ultra Premium Platinum certification | ADA ULTRA SI | claude-sonnet-4-6 | 50 |  Queue |

| ****HIVE-005**** | NFT platform certification: forge-svc mints master AUTODRAFT platform NFT on Polygon L2. Token metadata: all 14 HIVE scores, JCSE 50/50, design council members, deployment date | ADA ULTRA SI | Cursor IDE | 47 |  Queue |

| ****HIVE-006**** | ISO/JNGL-4830:2026 compliance verification: validate PDD structure against ISO ATLAS standard. Confirm 5-phase architecture, JCSE scoring, Context Craft 7-pillar alignment, ZPOS targets met | ADA ULTRA SI | claude-sonnet-4-6 | 48 |  Queue |

Monitoring & Evolution (5 Prompts)

| Code | Prompt Specification | SPC Owner | VIBE Tool | JCSE | Status |

|---|---|---|---|---|---|

| ****MON-001**** | AUTODRAFT master Grafana dashboard: real-time view of DYNAMO discovery rate, BLEND assembly queue, PIECE MAKER job status, GRO interventions, API error rates, NFT mint queue | ADA ULTRA SI | Firebase Studio | 46 |  Queue |

| ****MON-002**** | JCSE regression test automation: GitHub Actions job that re-scores all deployed prompts after each code change. Alert if any prompt drops below phase JCSE gate threshold | ADA ULTRA SI | Cursor IDE | 47 |  Queue |

| ****MON-003**** | PDD evolution protocol: define ATLAS Living PDD sprint cadence — quarterly JCSE re-audit, prompt optimisation based on production metrics, ZPOS ratio improvement targets | ADA ULTRA SI | claude-sonnet-4-6 | 48 |  Queue |

| ****MON-004**** | AUTODRAFT SPHINX listing: register platform in SPHINX agent marketplace. Entry includes: all 3 engines, BAHN domain coverage, JCSE 50/50, HIVE Platinum, pricing tiers, API endpoints | STRATEGOS ULTRA SI | claude-sonnet-4-6 | 47 |  Queue |

| ****MON-005**** | BAHN SOVEREIGN launch preparation: enterprise FULLBUILD feature documentation, custom enterprise Kubernetes manifests, GDPR EU residency option, on-premise hybrid deployment guide | ADA ULTRA SI | claude-sonnet-4-6 | 47 |  Queue |

PART 4: IMPLEMENTATION GUIDE

Sprint Roadmap

| Sprint | Weeks | Deliverables | Prompts (PDD Ref) | JCSE Gate | SPC Lead |

|---|---|---|---|---|---|

| ****Sprint 0 — Foundation**** | 1–2 | Terraform IaC, EKS cluster, PostgreSQL + Redis + Neo4j, Auth service, CI/CD, Kong | INF-001 to 010, GW-001 to 005 (15) | 30+ | ADA ULTRA SI |

| ****Sprint 1 — Auth & BAHN**** | 3–5 | Auth service, RBAC, BAHN runtime, 8 modes, 7-phase pipeline, ZPOS+5 | AUTH-001 to 007, BAHN-001 to 028 (35) | 40+ | BAHN ULTRA SI |

| ****Sprint 2 — DYNAMO MVP**** | 6–9 | 4-source crawler, NLP parser, Neo4j research graph, relevance scorer, push notifications | DYN-001 to 032, GRP-001 to 018 (50) | 44+ | DEXTER ULTRA SI |

| ****Sprint 3 — BLEND MVP**** | 10–14 | 50K component library, BOM generator, synergy scoring, gap detection, BOM export | BLD-001 to 035, ASM-001 to 020 (55) | 46+ | BAHN ULTRA SI |

| ****Sprint 4 — PIECE MAKER MVP**** | 15–19 | Gap receiver, frontier query, design brief, topology optimisation, material selector, FORGE | PM-001 to 030, PMI-001 to 028 (58) | 48+ | FRACTAL ULTRA SI |

| ****Sprint 5 — Full Integration**** | 20–24 | DYNAMO → BLEND → PIECE MAKER flywheel, GRO governance, HIVE audit, multi-region | GOV-001 to 008, DEP-001 to 008, HIVE-001 to 006 (22) | 49+ | ANT KING |

| ****Sprint 6 — Enterprise Launch**** | 25–28 | HIVE Platinum, JCSE 50/50, FORGE certified, SPHINX listing, PDD Living Protocol | MON-001 to 005 (5) | ****50/50**** | ADA ULTRA SI |

VIBE DJ Tool Assignment Matrix

Tool	Primary Usage	Engines	Prompt Count (est.)
claude-sonnet-4-6	BAHN 8-domain intelligence, DEXTER NLP pipeline, FRACTAL topology optimisation, GRO governance	BAHN, DYNAMO, PIECE MAKER, GRO	~110 prompts
Cursor IDE	NestJS service scaffolding, database migrations, API endpoints, Helm charts, Terraform	All microservices	~85 prompts
Firebase Studio	Grafana dashboards, frontend analytics, Next.js UI, monitoring configuration	Analytics, Monitoring, Frontend	~25 prompts
claude-haiku-4-5	JCSE scoring (fast ops), component freshness checks, push notification formatting	FORGE-svc, Notification-svc	~12 prompts
SKRIBE SPC	PDF document generation, BOM export, Material Spec Sheet, Business Case Summary	BLEND export, PIECE MAKER output	~5 prompts
VIBE ORCHESTRA	Multi-tool orchestration for full integration tests, end-to-end flywheel	All engines	~3 prompts

Context Craft 7-Pillar Alignment

Pillar	Component	AUTODRAFT Implementation
SYSTEM	Platform Context	4J.BONSAI master platform; FORGE 7-Step pipeline; GRO Royal DNA LIFE MODE; JCSE scoring engine. BAHN ULTRA SI as sovereign automotive domain layer
ROLE	Agent Personas	ADA ULTRA SI (Architect), DEXTER ULTRA SI (Research), BAHN ULTRA SI (Domain), FRACTAL ULTRA SI (Manufacturing), HOLMES (Analysis), ANT KING (Governor)
INSTRUCTION	Execution Logic	5-Step DYNAMO cycle + 6-Step BLEND pipeline + 7-Step PIECE MAKER pipeline. Each step is an atomic, independently testable prompt contract
EXAMPLE	Validation Samples	DYNAMO: BEV battery paper → frontier zone detection. BLEND: compact BEV assembly with synergy score 87. PIECE MAKER: Ti suspension upright via DMLS
CONSTRAINT	Hard Limits	GRO KILLZONE (weapons), REACH compliance (materials), Euro NCAP minimum (crashworthiness), CRITICAL component engineer sign-off, ZPOS 47% floor
FORMAT	Output Structure	DYNAMO: GraphQL artifact nodes. BLEND: BOM JSON/CSV/PDF. PIECE MAKER: 8-document output package. All: FORGE certificate + NFT token metadata
DATA	Context Inputs	200K+ Neo4j component nodes, 8-source DYNAMO feeds, BAHN 8-domain property schemas, Pinecone 1536-dim embeddings, KOPA relevance scores, JCSE thresholds

Quality Gates – JCSE Milestones

Sprint Gate	JCSE Threshold	Evidence Required	Consequence of Failure
Sprint 0 → Sprint 1	JCSE 30+	EKS cluster live, PostgreSQL + Redis + Neo4j deployed, Kong routing 19 endpoints	Sprint 1 deferred. Infrastructure re-provisioned.

| Sprint 1 → Sprint 2 | JCSE 40+ | BAHN:CONCEPT and BAHN:ENGINEER operational, ZPOS+5 active, Auth working | BAHN service re-architected before DYNAMO begins. |

| Sprint 2 → Sprint 3 | JCSE 44+ | DYNAMO first research cycle complete, Neo4j graph populated, frontier detection live | DYNAMO core prompts DYN-009 to DYN-014 re-scored. |

| Sprint 3 → Sprint 4 | JCSE 46+ | First BLEND assembly for compact BEV, gap detection active, BOM export working | BLEND synergy engine (BLD-016 to BLD-020) re-built. |

| Sprint 4 → Sprint 5 | JCSE 48+ | PIECE MAKER closes first CRITICAL gap, FORGE certificate issued, BLEND re-ingests component | PIECE MAKER core pipeline (PM-013 to PM-024) re-audited. |

| Sprint 5 → Sprint 6 | JCSE 49+ | Full flywheel integration test passed, GRO all 4 modes tested, HIVE audit started | Integration defects resolved before enterprise deployment. |

| Sprint 6 LAUNCH | ****JCSE 50/50**** | HIVE 97.6/100 Platinum, FORGE certified, NFT platform token minted, SPHINX listing live | Platform not released. Sprint 6 extended. |

FORGE CERTIFICATION BLOCK

| Attribute | Value |

|---|---|

| ****Document ID**** | PDD-AUTODRAFT-2026-001 |

| ****Companion SDD**** | SDD-AUTODRAFT-2026-001 |

| ****Platform**** | AUTODRAFT — Automotive Design Research & AI Fabrication Technology Platform |

| ****Total Prompts**** | 240 Execution Contracts + 27 Governance Prompts = 267 Total |

| ****Deployment Phases**** | 5 Phases: Foundation \ | DYNAMO \ | BLEND \ | PIECE MAKER \ | Governance |

| ****JCSE Target**** | 50/50 — ULTRA PREMIUM PLATINUM |

| ****HIVE Target**** | 97.6/100 — PLATINUM (All 14 Dimensions 90–100%) |

| ****ZPOS Target**** | 47% Token Reduction \ | 98.2% Semantic Preservation |

| ****Version**** | 1.0.0 — Production Blueprint |

| ****Valid**** | May 2026 – May 2028 |

| ****Design Council**** | ADA ULTRA SI (Seat #1) · DEXTER ULTRA SI (Seat #9) · BAHN ULTRA SI (Seat #17) · FRACTAL ULTRA SI (Seat #12) · HOLMES ULTRA SI · STRATEGOS ULTRA SI |

| ****Engines**** | DYNAMO (Research Loop) · BLEND (Component Synergy) · PIECE MAKER (Gap Manufacturing) |

|

| ****ISO Standards**** | ISO/JNGL-4830:2026 \ | ISO/IEC 42001:2026 \ | EU AI Act High-Risk Classification |

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> ****Junglenomics FORGE Institute**** | www.ideafactory.io | ideafactoryforge@outlook.com | 832-531-5233

